AMAZON BASE BUILDING COMPLIANCE STANDARD

Task Title	Lightning protection
APM Object	CT.ELEC.LIGHTNING
APM Description	Lightning protection systems.
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure. CT evidence must be additionally uploaded to the DMS (WorkDocs in DE) To suggest improvements please raise a SIM or contact document authors Leonardo Vitale (LEOVITAL) Rob Gisby (EWARTGXR) or APM, Benito Sanchez Sanchez de Puerta (BPPUERTA) SIM - Link to SIM template

Last Update: (Version)	abr-24	**Last Update Change Detail:**	PM evidence changed from 3M to 12M.	**Activities may be covered by No-Go restrictions:**	Not Specified	**Special Certification Required:**	**Delivered By:**
Evidenced part or full by PM:	Not Specified	**Related BBM PM:**	BLDG.LIGHTNING	**Related CT Areas:**	CT.ELEC.ELECINST - CT.ELEC.EMERGENCYAC - CT.ELEC.POWERGENERATOR - CT.ELEC.UPS		

Scope

Any site with an electrical installation. Inspection of the earthing and electrical discharging systems is mandatory as part of the inspection of the electrical installation as a whole.
More than 36 kV - high voltage
However, EU RME simplifies to LV < 1 kV AC and HV ≥ 1 kV AC.
Any site where an emergency lighting system (including battery) is installed to aid safety in the event of power failure and/or safe evacuation of the site in emergency situations.
Inspection requirements of electrical installations are covered by "Compliance HV inspection and test", "High/Medium Voltage" and "Low Voltage" Standards.
Any site with an electrical installation. CEN and CENELEC standards define incoming electricity at:
Less than 1 kV - low voltage installations
Between 1 kV and 36 kV - medium voltage installation
Any site where diesel power generators are commissioned to produce electricity, generally as a back-up in emergency situations. This includes all diesel generators, whether stationary or mobile.
This Compliance Standard only covers the inspection requirements associated with the electrical components of diesel generators. The maintenance of a UPS is covered in CT.ELEC.UPS.

Using this document

This document is intended as a guide and may not be exhaustive. Whilst this document is based extensively on professional guidance and subject to periodic review, individual sites must be diligent to ensure they are aware of, and comply to, local and changing regulations.
The format of these standards is intended to improve navigation between maintenance and compliance standards and to allow frequent, minor updates to a single living resource. Please avoid offline or printed versions and check version dates and change log.
In the event of becoming aware of errors or omissions in this document, sites must provide detail, including references to relevant source information to their local planner for escalation - SIM Link above.
This document is considered released as draft.
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Maintenance and Compliance Relationship

Where relevant, associated Base Building PM's are listed on the header . In many cases maintenance contributes to, or fulfils compliance requirement. This may not be exhaustive and is intended to assist users to identify relevant maintenance.
If a BBM Maintenance activity fully meets the compliance requirements below, periodic upload of PM evidence will be required to APM and DMS / Workdays depending on region.

Key to Abbreviations

The key to the below document is as follows:
CT - Amazon compliance / statutory task - not otherwise covered by a BBM Standard PM. Evidence to be provided into APM and DMS upon task closure.
CT* - This is a conditional task based or based on findings from previous inspections. If a frequency is not used, a further appropriate frequency needs to be in either the region, or 'ALL'. Local teams must ensure appropriate frequency.
(CT) - Compliance task which is likely met or exceeded by a CT task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met.
PM - Requirement can demonstrated through appropriate maintenance - Refer to relevant PM Standard
(PM) - Maintenance task which is met or exceeded by a PM task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met including regional / manufacturer requirements.
EVID - Periodic upload of evidence to DMS, for non-CT task lines. For example, evidence of PM completion, certification, licence etc.
EVID* - Conditional requirement to upload evidence at the interval provided
N A R - No Additional Requirement

Required Tasks

Region	Task Index	Task Frequency	Task Requirements	Qualification of Inspector	Records Required	PM	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	Additional
All	1	In accordance with manufacturer's instructions and recommendations	Regular maintenance and cleaning of lightning protection systems and associated infrastructure.	Competent person (Accredited Electrician on the Electrical Safety Register e.g. NICEIC, ECA, SELECT)	Maintenance logbook	PM					EVID		
All	2	At least every 12 months. More frequent inspections might be required based on risk assessment and operating conditions.	Regular thorough inspections to assess the integrity and operation of the lightning protection systems (e.g. visible damage due to lightning strikes, integrity of all elements, continuity of conductors, corrosion, etc.)	Competent person (Accredited Electrician on the Electrical Safety Register e.g. NICEIC, ECA, SELECT)	Inspection records. Risk assessment	(PM)					EVID		

Related Regional Legislation

Region	Relevant legislation	Vendor Certification	Statutory requirement
CZ	Decree 73/2010 Coll., Decree. 50/1978 ČSN 331500, ČSN 332000-6 ed. 2, ČSN EN 62305, ČSN 332570, ČSN 332000, , ČSN 331600.ed.2,	Competent, trained review technician	Light conductor checks carried out every 5 years before 2009 and post-2009 inspections for 4 years. In hazardous areas, the frequency is reduced to 2 years. Visual inspection to be carried out between checks (1 or 2 years) Frequency determined by the Protocol on the determination of external influences, 6 and 60months.
DE	VdS-Richtlinie 2010 - Risikoorientierte Blitz- und Überspannungsschutz DIN EN 62305-3 VDE 0185-305-3 – Beiblatt 3 - Zusätzliche Informationen für die Prüfung und Wartung von Blitzschutzsystemen Betriebssicherheitsverordnung (BetrSichV) Arbeitsstättenverordnung (ArbStättV) §4 (3) Respective law of the federal state with reference to the building permit	Qualified electrician and lightning protection expert (Blitzschutz-Fachkraft). Additional inspection: expert according to building law, certificate necessary.	Before initial commissioning, after extensions or modifications to the building, after lightning strikes. Protection classes I and II require visual inspections yearly and extended inspection every 2 years, while classes III and IV require visual inspection every 2 years and extended inspections every 4. More frequently in case of a critical event or based on risk assessment. Additional inspection based on building law.

ES	UNE 21.186: 2011 standard DB- SUA8 – CTE (Construction Technical Code- Safety Using Basic Document number 8 – lightning protection).	Qualified technician.	Regular maintenance. Inspections conducted at least every 12 months.
FR	Ministerial Order of October 4th, 2010 Article 16 to 23 https://www.legifrance.gouv.fr/loda/id/JORFTEXT000023081900/	Qualified organism	Visual inspection every year and full inspection every 2 years.
IT	Legislative Decree 8/04/2008 n.81 DPR 462/01 DM 22/01/2008 n.37 CEI EN 50110-1 (Operation of electrical installations) CEI 11-27 (Work on electrical installations) CEI 81-2 (Guidelines for lightning protection systems inspection) CEI EN 62305-4/EC (Maintenance)	National authorities (ASL, ARPA) or accredited inspection organization.	Regular maintenance. Inspections every 5 years after putting into service, or every 2 years if the building is at risk of fire according to DPR 155/11.
PL	Act of 7 July 1994 on Construction Law (Journal of Laws 2020, item 1333) Regulation of Minister of Infrastructure of 12 April 2002 on technical conditions to be met by buildings and their location. (Journal of Laws 2019, item 1065)	Qualified or competent person. Authorised by Construction Law.	General inspection conducted every 12 months. Testing every 5 years.
ROI	Safety, Health and Welfare at Work (General Application) Regulations 2007 The NSAI - IS 10101:2020 on 6 March 2020	Competent person approved by RECI or ECSSA.	Test and inspection for lightning and earth protection at least ever 12 months. Recommended every 11 months to cover every month over a 12 year period
SK	Act No. 124/2006 Coll. on safety and health at work Ordinance No. 508/2009 Coll. on details for ensuring safety and health protection at work with pressure, lifting, electrical and gas technical equipment Ordinance No. 94/2004 Coll. laying down technical requirements for fire safety in the construction and use of buildings	Qualified and certified person	Protection systems level I and II require a visual inspection yearly and a functional inspection and testing every 2 years. Protection systems level III and IV require visual inspection yearly and a functional inspection and testing every 4 years.
UK	Health and Safety at Work etc. Act 1974 (legislation.gov.uk) The Electricity at Work Regulations 1989 (legislation.gov.uk) BS 7671 / IET Wiring Regulations Health and Safety at Work (Northern Ireland) Order 1978 (legislation.gov.uk) The Electricity At Work Regulations (Northern Ireland) 1991 (legislation.gov.uk) BS EN 62305-3 Protection Against Lightning BS 7430 Code of Practice for Earthing	Electrician registered on the Electrical Safety Register. Approved by NICEIC, ECA or SELECT.	Test and inspection for lightning and earth protection at least ever 12 months. Recommended every 11 months to cover every month over a 12 year period.